FlowLine

SEWERAGE / DRAINANGE





UPVC

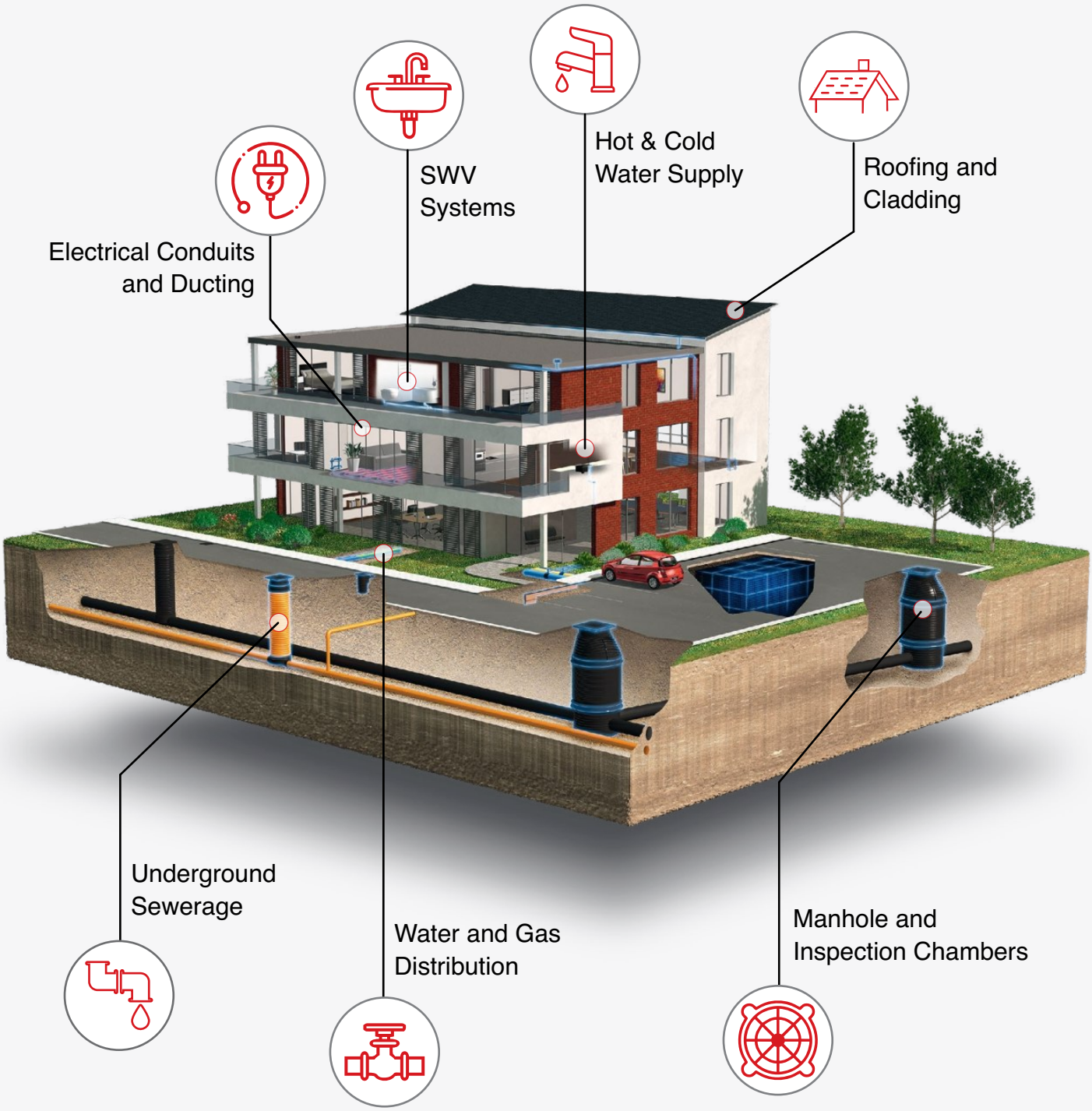
SPECIALLY DESIGNED FOR
UNDERGROUND SEWERAGE AND DRAINAGE

magnaplast

DADEX

DADEX

Leader in Roofing and Thermoplastic Piping Systems



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www.dadex.com

ABOUT DADEX

Dadex was established as a public limited company in 1959. We started with the manufacturing of FC roofing systems at our facilities in Hyderabad, under the dynamic leadership of our Founder Chairman, Mr. Kassim Dada. Being a forward-looking organization, his son, Mr. Sikander Dada kept up to date with the market, and was instrumental in the diversification with plastic piping systems. Presently we produce various thermoplastic systems such as UPVC, PP-R, PP-RCT, PP, PE.

Headquartered in Karachi, our two manufacturing facilities based in Hyderabad and Sundar Industrial Estate (near Lahore) are equipped with state of the art machinery. Our joint efforts have led Dadex to the ISO 9001, ISO 14001, ISO 45001 and PSQCA certifications.

We aim to set standards in every field we enter. Customer focus, quality and innovation are reflected in every aspect of our business. Offering customized solutions to a cross section of markets for over 66 years, Dadex today is a hallmark of trust and reliability.

We believe that our strength lies in delighting our customers. That is why providing quality products backed by round the clock technical support services are the Dadex promise to all our valued customers.

EXTENSIVE PRODUCT RANGE

Water Supply - Infra-Structure

- Aquadex (UPVC Pressure Pipe System)
- T-Flex (PE 100 Pipes & Fittings)

Water Supply - Housing

- Polydex Premium (PP-RCT Pipe System)
- Polydex (PP-R Pipe System)
- Thermoline (PP-R Pipe System)
- Aquadex (UPVC Pressure Pipe System)

Soil, Waste and Vent (SWV)

- Nikasi - Solvent (UPVC Pipe System)
- Nikasi - Rubber Ring (UPVC Pipe System)
- Polyduct (Low Noise System)

Underground Sewerage and Drainage

- Flow Line (UPVC Underground Sewerage System)
- Inspection Chambers and Manholes

Cable Ducting

- PE Cable Duct Pipes
- Electroduct (UPVC Conduits as per IEC - EN Standard)
- Conduits (UPVC Conduits as per Dadex Standard)

Gas Supply

- T-Flex for gas (PE 80 and PE 100)

Introduction

Flowline piping system is a suitable solution for underground drain and sewer applications in homes, commercial plazas, community buildings and infrastructure projects. This easy push-fit rubber ring/solvent cement jointing system is durable, corrosion free and lightweight. Flowline is the first UPVC based underground sewer pipe system in Pakistan that conforms to international quality standards. It is specially designed for Underground Sewerage and Drainage Applications.



Material

Flowline pipe system is manufactured from Unplasticised Polyvinyl Chloride (uPVC) compound. Flowline pipes and fittings are terra cotta (orange brown) in color, which is recommended color for buried drainage system.

Standards and Specifications

Flowline pipes and fittings are manufactured in accordance with EN 1401-1.

Available Range

Flowline pipes and fittings are available from 110mm to 500mm.
A range of imported injection molded fittings could be made available.
The pipes are available in standard lengths of 4 & 6 meters and the sizes over 250mm are available in Bell End only.

Nominal Outside Diameter (mm)	Mean Outside Diameter min-max (mm)	Min Wall Thickness (mm) SN 8, (SDR 34)	Min Wall Thickness (mm) SN 4, (SDR 41)
110	110.0 - 110.3	3.2	3.2
160	160.0 - 160.4	4.7	4.0
200	200.0 - 200.5	5.9	4.9
250	250.0 - 250.5	7.3	6.2
315	315.0 - 315.6	9.2	7.7
355	355.0 - 355.7	10.4	8.7
400	Available by import		
500	Available by import		

(Wall thickness and outside diameter refer to pipes only)

Features & Benefits

- Efficient Disposal: Exceptionally low friction and smooth inside surface of Flowline minimizes the build up of deposits commonly seen in conventional sewer pipe systems.
- Low Bacterial Growth: All conventional sewer pipes collect a coating of algae or slime. Flowline offers lesser grips for slime.
- Thermal Expansion/Contraction: Flowline pipe system with rubber ring socket joint allows natural provisions against thermal expansion.
- Non Corrosiveness and chemical Resistance: One of the most remarkable features of Flowline is its non-corrosiveness. It resists chemical reactions from acids, alkalis and salt solutions.
- Lightweight and Easy to install: Flowline pipes are light weight and easy to install. This leads to lower transportation and installation costs.
- Leak Free: Flowline has water tight connections. There is no chance of infiltration or pollution of ground water.
- Easy to Maintain: Flowline pipe systems are easy to access after installation for inspection, cleaning, repairs, etc.

Laying & Installation

The following information is based on the recommendations of BS 5955 part 6 “code of Practice for installation of UPVC pipe work for gravity drainage & sewer” BS EN 1610 “ Construction and testing of drains and sewers”

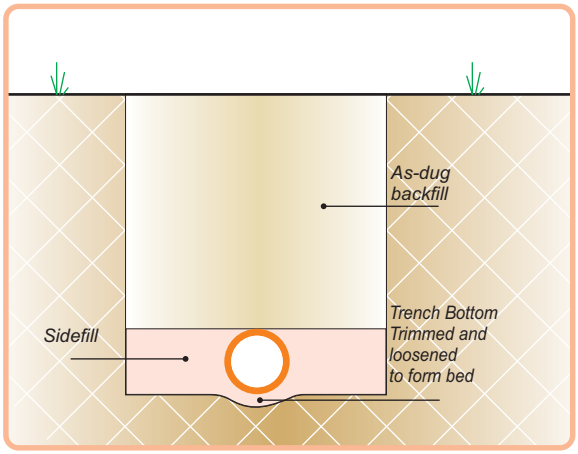
Bedding

A. Drain pipes laid on trench bottom

Where the 'as-dug' material is suitable*, the bottom of the trench may be trimmed to form the pipe bed. (*Suitable material is defined as granular material).

Small depressions should be made to accommodate sockets. After the pipe has been laid these should be filled carefully ensuring that no voids remain under, or around the sockets.

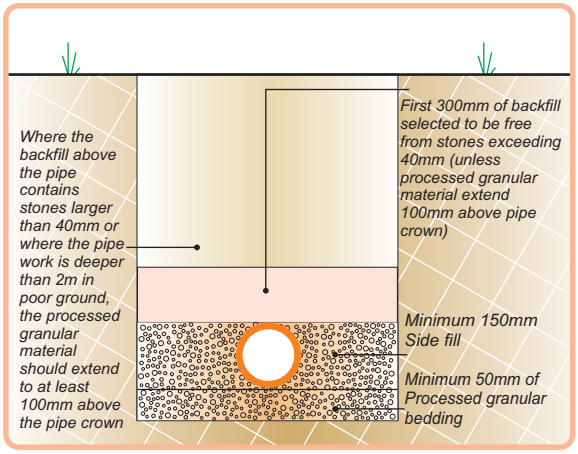
When the formation is prepared, the pipes should be laid upon it true to line and level within the specified tolerances. Each pipe should be checked and any necessary adjustments to level made by raising or lowering the formation, ensuring that the pipes finally rest evenly on the adjusted formation throughout the length of the barrels. Adjustment should never be made by local packing.



A - Pipe Laid on Trench Bottom

B. Drain pipes laid on a 50mm minimum granular bed

When the as-dug material can be hand trimmed by shovel and is not puddled when walked upon, a 50mm depth of bedding material may be used. In this case the material must be nominal 5 to 9mm size aggregate with no sharp edges, i.e. Pea gravel.



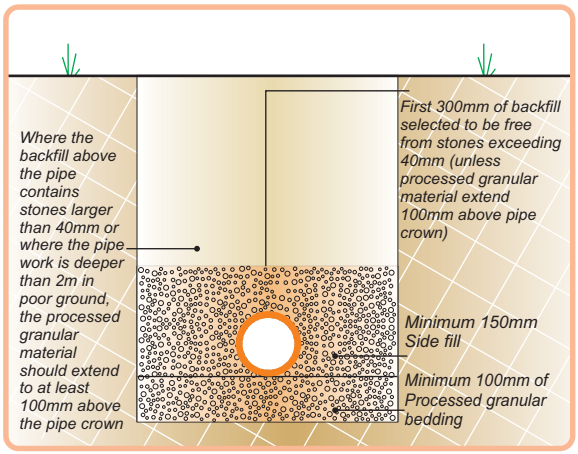
B - Pipes on 50mm Granular Bed

C. Drain pipes laid on a 100mm minimum granular bed

When the as-dug material is not suitable as a bedding, a layer of suitable granular material must be spread evenly on the trimmed trench bottom before the pipes are installed.

The trench should be excavated to allow for a minimum thickness of 100mm granular bedding under the pipes. The trench formation should be prepared, the bedding placed and the pipes laid in accordance with recommended practices.

When the pipes are to be laid on rock, compacted sand or gravel should be used, or in very soft or wet ground, the bedding should be as detailed above.



C - Pipes on 100mm Granular Bed

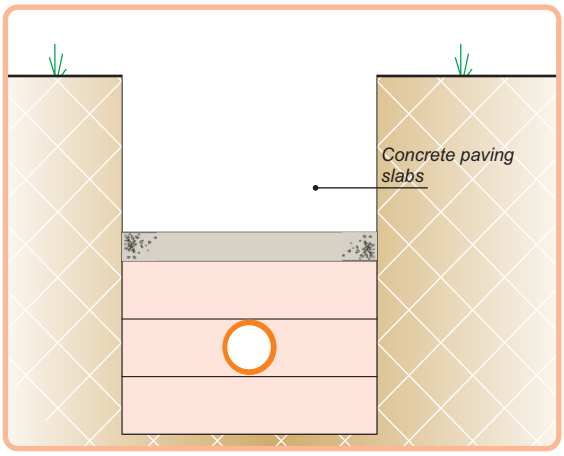
Backfill Sequence

- Place suitable sidefill material evenly on each side of the pipe in 100mm layers. Pay particular attention to the area under the lower quadrants of the pipe. Hand tamp well at each layer up to the pipe crown. Leave the pipe crown exposed.
- If 'as-dug' material is free from stones exceeding 40mm, imported processed granular material is not needed above the pipe crown. Cover the pipe crown with a minimum of 300mm of compacted 'as-dug' material. If 'as-dug' material contains stones larger than 40mm, or the pipe is deeper than 2 metres in poor ground, extend the processed granular material for at least 100mm above the pipe crown.
- In both cases, hand tamp the material fully at the sides of the pipe while tamping lightly over the crown. Continue hand tamping until a finished layer of 300mm, (225mm in adoptable situations), has been placed over the pipe.

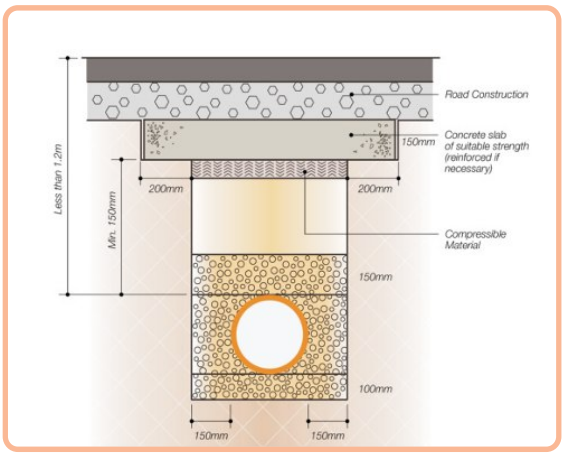
- 'As-dug' material may be backfilled in 300mm/225mm layers and mechanically tamped. Dumpers or other vehicles must not be driven along the pipe tracks as a means of compacting. Surround vertical or steeply raking pipes with 150mm bedding material, suitably tamped up to the invert level of the incoming pipe (Backdrops) or to ground level.

Pipe Protection

As uPVC pipes are flexible they can accommodate a degree of ground movement and pressure without damage. However, if the pipe needs protection the following recommendations should be followed:

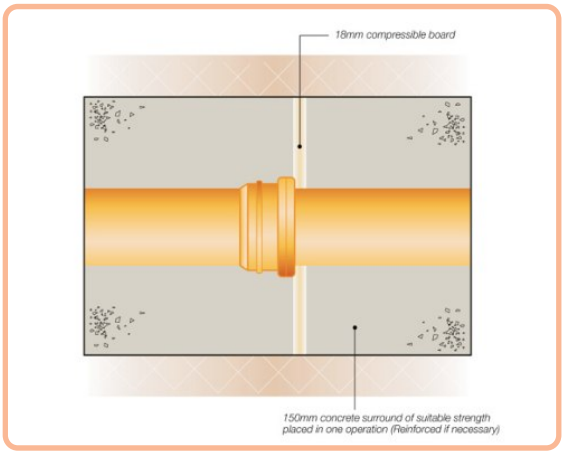


Traffic Free Area: In areas where no loading is expected (e.g. in gardens) pipes at depths less than 0.6 metre, should, where necessary, be protected against risk of damage from garden implements, for example by placing over them a layer of concrete paving slabs with at least a 75mm layer of suitable material between pipe and slab.



Public highways / adoptable situations: In areas where loading is expected, pipes laid at depths less than 0.9 metre below the finished surface of a road, (1.2m in adoptable situations) should be protected with a concrete slab of suitable strength which should bridge the full width of the trench so it sits on the trench wall. or, alternatively the pipe can be totally surrounded in concrete. Concrete of suitable strength or the requirement for reinforced concrete to be determined by the engineer or adopting authority. The normal maximum depth for all installations is 10 metres.

Use of concrete: If pipes are to be surrounded with concrete, make sure they do not float when the concrete is poured. Filling the pipes with water will generally provide enough ballast but side restraint May be needed to maintain alignment.
To maintain certain degree of flexibility, insert 18mm compressible material, such as fibreboard or polystyrene, around the pipe joints. These boards must be at least the width of the concrete surrounds.



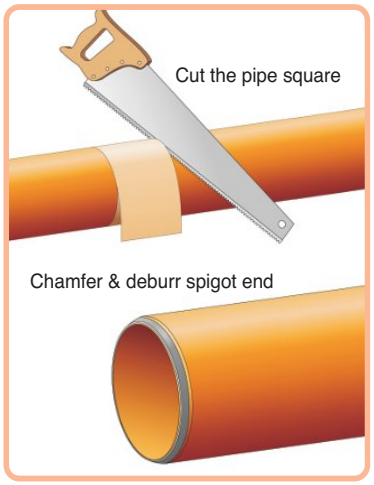
Pipes under buildings: A drain may run under a building if at least 100mm of granular or other flexible filling is provided round the pipe. On sites where excessive subsidence is possible additional flexible joints may be advisable or other solutions such as suspended drainage. Where ground settlement is expected and the crown of the pipe is within 300mm of the underside of the slab, concrete encasement should be used integral with the slab.

Pipes penetrating walls: Where a short length of pipe is to be built into a structure, a suitable wall protection sleeve should be used. The short length of pipe should then be inserted throw the wall protection sleeve, and fixed with couplers placed either side within 150mm from the wall face. The length of the next 'rocker' pipe should not exceed 0.6 metre. This will compensate for any settlement of the building or made up ground.
Alternatively, where it is not necessary for a pipe to be built into a structure, the provision of a lintel, relieving arch or sleeve may be used, leaving a gap of not less than 50mm around the pipe. Effective means should be adopted to prevent the entry of gravel, rodents or grasses.

Pipe Cutting and Jointing

Preparing Pipe Ends

Pipes cut on site must be clean cut at right angles to their horizontal axis.
De-burr the cut end with a scraper if the cut end is to be inserted into a ring-seal or push-fit joint. Chamfer the spigot end: this is essential to ensure that the sealing ring is not displaced during insertion.

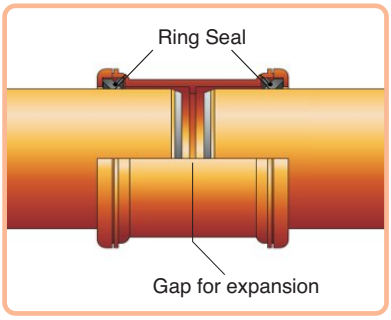


Depth of Entry Mark

Some plain ended fittings have a depth of entry mark moulded on the spigot. This depth of entry allows the pipe to expand into the fitting socket by a minimum of 12mm. Insert the spigot into the socket until the depth of entry mark is just visible. All pipes (whether site cut or otherwise) and other plain ended fittings must be inserted to the full depth of the socket, marked at the socket face and then withdrawn at least 12mm.

Ring-Seal / Push-Fit Jointing

- Ensure any pipe cut on site is also chamfered.
- Check that the sealing ring is properly seated in its housing in the socket of the fitting.
- Ensure all components to be joined are dry, clean and free from grit or dust. Note any deep scratches on the pipe or fitting spigot as these may prevent the sealing ring from forming a water tight seal.
- Lubricate evenly around the pipe or fitting spigot end with Silicone Lubricant. Do NOT lubricate inside the socket and not the ring seal. The spigot can then be inserted into the socket. Use of washing liquid is not recommended as lubricant. However, soap solution can be used as lubricant.
- Correctly align the components to be joined.
- Push the pipe or fitting spigot fully into the socket. Mark the pipe or fitting spigot at the socket face and then withdraw it by a minimum of 12mm to allow for thermal expansion.
- Make a subsequent check to ensure that the expansion gap is not lost during further installation work.



Solvent Cement Jointing

- Before using solvent based cleaners or cement:
  - Read instructions on the can
  - Ensure there is sufficient ventilation.
- Make sure pipe or fitting spigot and solvent weld socket are dry, clean and free from grit or dust.
- Clean surfaces of spigot and socket with Dadex Degreasing Cleaner. Apply liberally using a clean non synthetic rag or absorbent paper.
- Apply one coat of Dadex Solvent Cement. Apply an even coat to both surfaces using the applicator provided or a paint brush. Stroke the cement along and not around the surfaces.
- Immediately insert pipe or fitting spigot fully into the socket. Each solvent weld joint MUST be completed within 1½ minutes.
- Hold for 20-30 seconds. Remove any surplus cement from the mouth of the socket.
- The joint may be handled after 10 minutes and commissioned after 24 hours.

Safety

When making solvent weld joints it is essential to observe normal safety rules for handling solvent:

- Never smoke or bring naked flames near the area of work
- Work in a well ventilated area to avoid inhaling fumes
- Close the solvent container after use and store in a cool area
- Do not allow solvents or cleaners to come into contact with skin.

Transport

Generally pipes are delivered on flat bed trucks.

- When vehicles with a flat bed are used for transporting loose pipes, make sure the bed is free of nails and other projections.
- Support pipes throughout their length. Load pipes so that they do not overhang the vehicle by more than one Meter. Always load pipes with larger diameters and thicker walls before those of smaller diameters and thinner walls.
- Flowline pipes should always be lifted off the vehicle, not dragged, thus avoiding damage to the pipe ends. Make sure vehicles have adequate side supports at approximately 2 metre spacings, and that all uprights are flat, with no sharp edges. Secure pipes during transit.
- Fittings are supplied in cardboard boxes or plastic bags.

Handling

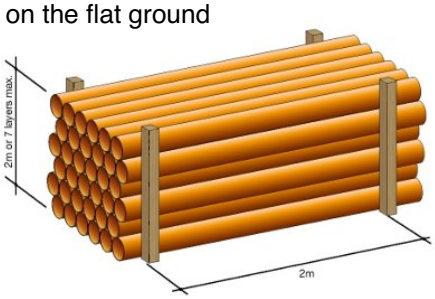
- Always be careful to avoid damage when handling pipe. Cold weather reduces their impact strength, so take extra care when handling pipe in cold conditions.
- Load and unload loose pipes by hand and avoid using skids. When loose pipes have been transported one inside the other, always remove the inner pipes first.
- Do not drop or drag pipes.

Storage

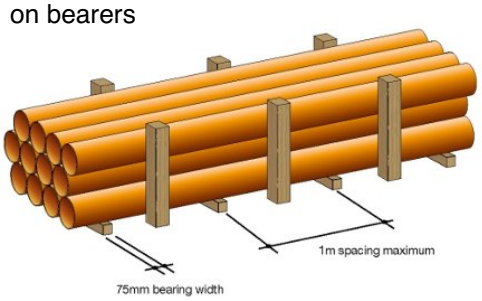
Loose Pipes

- Store loose pipes on a reasonably flat surface free of sharp projections. Provide side supports at least every 2 metres. These supports should preferably consist of battens at least 75mm wide.
- Ideally, loose pipes should be uniformly supported throughout their entire length. If this is not possible, place timber supports at least 75mm wide at 1 metre maximum centres beneath the pipes.
- Stack pipes of different size and wall thickness separately. If this is not possible, stack pipes with larger diameters and thicker walls under those with smaller diameters and thinner walls.
- Do not stack pipes more than seven layers in height or above a maximum height of 2 metres.

Storage of loose pipes on the flat ground



Storage of loose pipes on bearers



Fittings

- Store fittings supplied in plastic bags away from direct sunlight.
- If fittings have to be stored outside in their plastic bags, open the bags to prevent a build-up of temperature.
- The above storage requirements apply to the normal climatic conditions. In hot areas reduce the stack height and store pipes and fittings under cover or in the shade.

Rubber Seal Rings

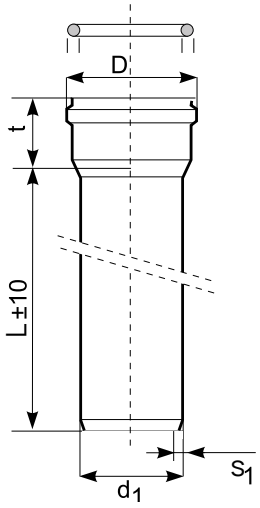
Make sure that rubber seal rings are kept in polyethylene bags or cartons in shade away from direct sunlight.



PRODUCT CATALOGUE

PIPES WITH SOCKET





Class B - SN4

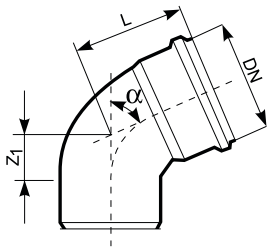
DN	d_1 [mm]	s_1 [mm]	D [mm]	t [mm]	L
160	160	4.0	183	110	1000
160	160	4.0	183	110	2000
160	160	4.0	183	110	3000
160	160	4.0	183	110	6000
200	200	4.9	226	120	1000
200	200	4.9	226	120	2000
200	200	4.9	226	120	3000
200	200	4.9	226	120	6000
250	250	6.2	287	140	3000
250	250	6.2	287	140	6000
315	315	7.7	355	160	3000
315	315	7.7	355	160	6000

Class C - SN8

DN	d_1 [mm]	s_1 [mm]	D [mm]	t [mm]	L
110	110	3.2	128	76	1000
110	110	3.2	128	76	2000
110	110	3.2	128	76	3000
110	110	3.2	128	76	6000
160	160	4.7	183	110	1000
160	160	4.7	183	110	2000
160	160	4.7	183	110	3000
160	160	4.7	183	110	6000
200	200	5.9	226	120	1000
200	200	5.9	226	120	2000
200	200	5.9	226	120	3000
200	200	5.9	226	120	6000
250	250	7.3	287	140	2000
250	250	7.3	287	140	3000
250	250	7.3	287	140	6000
315	315	9.2	355	160	2000
315	315	9.2	355	160	3000
315	315	9.2	355	160	6000

BENDS

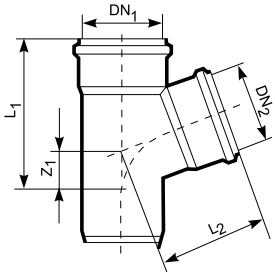




DN	angle	z_1 [mm]	L [mm]
110	15°	8	84
110	30°	14	85
110	45°	29	85
110	67°	40	104
110	87°	69	123
160	15°	32	88
160	30°	40	135
160	45°	47	135
160	67°	96	170
160	87°	64	142
200	15°	26	133
200	30°	49	160
200	45°	65	160
200	67°	68	168
200	87°	105	245
250	15°	19	170
250	30°	37	189
250	45°	57	189
250	87°	132	283
315	15°	23	198
315	30°	47	221
315	45°	72	221
315	87°	166	340

BRANCHES

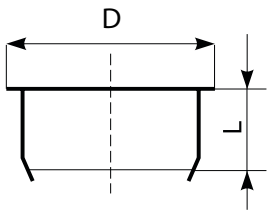




DN1/ DN2	angle	z_1 [mm]	L_1 [mm]	L_2 [mm]
110/110	45°	27	226	201
160/110	45°	16	244	252
160/160	45°	38	317	283
200/110	45°	17	280	271
200/160	45°	20	337	327
200/200	45°	46	402	359
250/110	45°	30	394	294
250/160	45°	33	472	396
250/200	45°	48	397	396
250/250	45°	50	560	530
315/110	45°	34	371	407
315/160	45°	15	415	395
315/200	45°	5	457	438
315/250	45°	15	650	515
315/315	45°	45	645	600
110/110	87°	55	186	142
160/110	87°	83	240	178
160/160	87°	116	297	213
200/110	87°	109	304	166
200/160	87°	143	352	216
200/200	87°	144	381	230
250/110	87°	62	317	256
250/160	87°	88	388	285
250/200	87°	107	422	310
250/250	87°	115	439	345
315/110	87°	67	352	266
315/160	87°	90	415	315
315/200	87°	100	450	340
315/250	87°	115	480	380
315/315	87°	135	525	430

PLUG

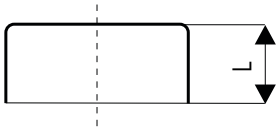




DN	L [mm]	D [mm]
110	32	120
160	42	170
200	50	217
250	80	265
315	80	333

END CAP

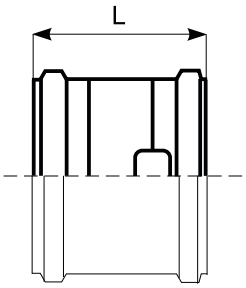




DN	L [mm]
110	43
160	52
200	64
250	68
315	77

SLEEVES

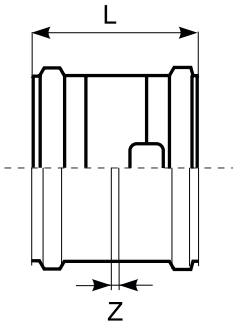




DN	L [mm]
110	105
160	138
200	215
250	250
315	293

DOUBLE SOCKETS

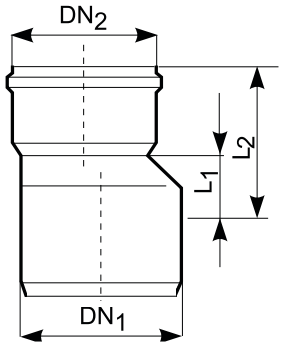




DN	Z [mm]	L [mm]
110	3	105
160	3	138
200	5	215
250	7	250

REDUCERS

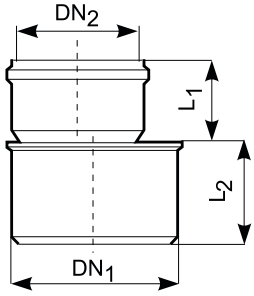




DN1/DN2	L_1 [mm]	L_2 [mm]
160/110	46	118
200/160	48	143
250/200	147	264
315/250	214	342

MULTI-STAGE REDUCTIONS

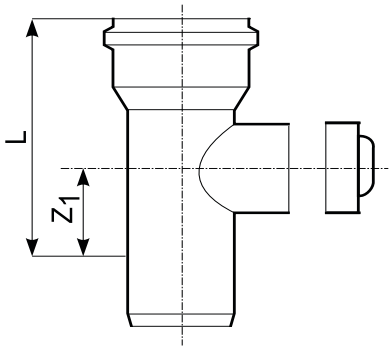




DN1/DN2	L_1 [mm]	L_2 [mm]
250/160	113	130
315/160	113	141
315/200	118	147

ACCESS PIPES - LOCKING PLATE

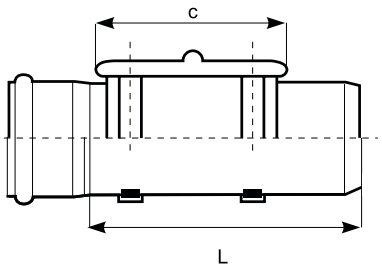




DN	Z1 [mm]	L [mm]
110	58	179
250	128	722

ACCESS PIPES - COVER WITH BOLTS

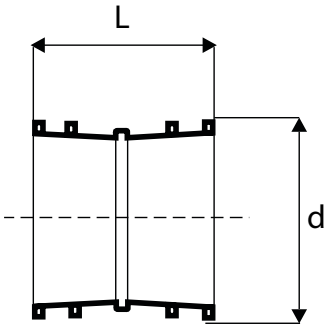




DN	c	L [mm]
110	221	279
160	282	332
200	394	494

WALL SLEEVES





DN	L [mm]	d [mm]
110	110	131
160	110	184
200	110	226
250	110	286
315	110	354
400	110	440
110	240	137
160	240	190
200	240	231.5
250	240	290
315	240	359
400	240	448
500	240	554









NON-RETURN VALVES

Introduction

We could blame it on climate change or on the “irresponsibility” of men, but it is a fact that more and more often, the rains that hit our country cause considerable problems of flooding in houses/buildings. The sudden and massive fall of several millimeters of water does not find a correct outlet and this generates water returns inside the buildings. Very often, despite the knowledge of the reasons that cause these problems, it is not possible to radically intervene on the sewer network and, therefore, it becomes necessary to adopt a solution directly on the building to be protected from flooding risks. This solution is called Non-return valve.

What is it and what are the causes of reflux

Knowing the principle of “communicating vessels”, according to which water tends to maintain the same level in all branches of a hydraulic circuit, when we talk about reflux, we define the maximum water level of a hydraulic circuit beyond which the leaking from the circuit itself. For this reason, if the sanitary utilities are below the ebb level (basically basements, cellars, garages, but also living quarters that are in any case below the level of the road level), flooding can occur during overload of the sewer network.

The reasons for the reflux effect may be the following:

- An incorrectly designed drainage system.
- High hydraulic flows caused by short but very intense rainfall.
- Increase in the volumes of rain channeled by collectors in relation to the building increase and urbanization.
- Water overload due to contemporaneity peaks in areas with high urban density.
- Malfunctions and occlusions downstream of the drainage system.

Material: Unplasticised Polyvinyl Chloride (uPVC)

Standard: EN 13564-1: 2003

Available Sizes: 110mm, 160mm and 200mm

Main Features

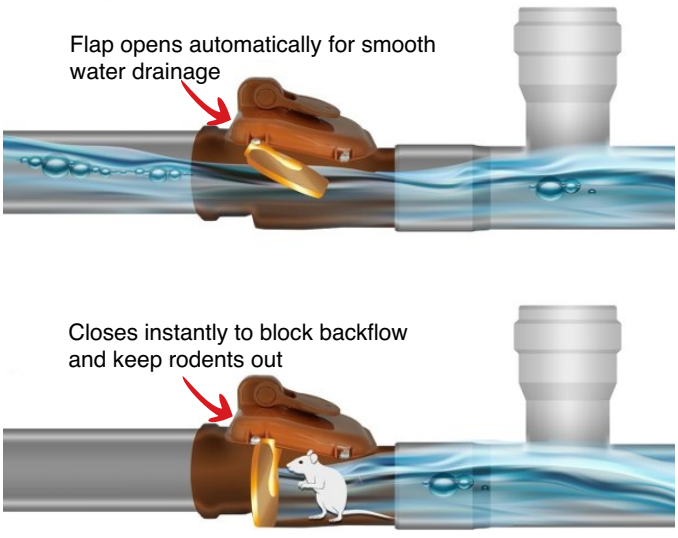
- High wall smoothness
- Very low weight
- Simple and quick assembly
- Possibility and ease of combining with other materials
- Total resistance to general and pitting corrosion
- High durability, estimated for 100 years
- Resistance to harmful effects of chemical compounds
- Resistance to deposits on the inner surfaces of pipes



The non-return valve as a protection and safety solution for homes

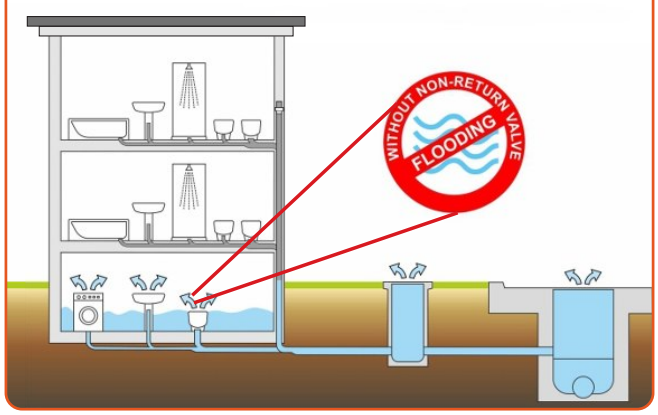
The non-return valve is designed for use within sewerage and drainage systems to eliminate the risk of flood damage caused by the backflow of sewage or flood water into properties through low-level entries, such as drain gullies and toilets. It is a reliable solution for protecting against the backflow of sewage and rainwater.

In addition to effective water flow control—ensuring that water only flows in one direction and minimizing the risk of flooding during heavy rainfall—the valve also acts as a barrier against rodents, preventing them from entering properties through the drainage system.

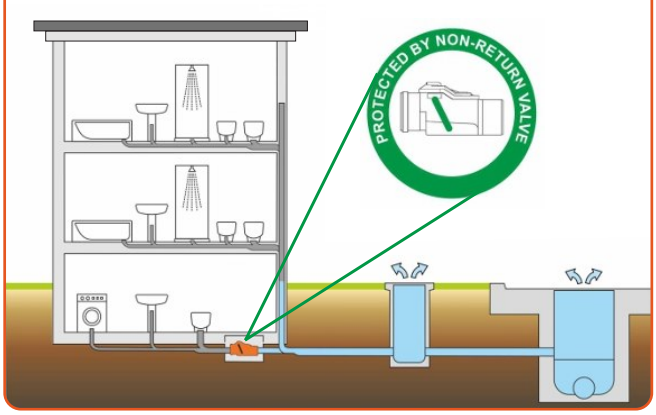


The non-return valve is housed in a suitable accessible and inspectable well. Its role is to create an effective barrier to any return of water flow coming from the sewerage system during any possible overload event. The non-return valve is equipped with a stainless steel plate that pivots in the direction of the water flow from the building to the sewer system. In the case of an ebb, the plate closes mechanically with the thrust of the water. The plate, acting on a special lever, can be permanently hermetically locked. Furthermore, its use is absolutely recommended in existing buildings, where it is possible to intervene on the connection piping to the sewer network and there is the physical space to create a pit that houses the valve. The non-return valve must be approved in accordance with the European standard EN 13564-1: 2003 which guarantees its requirements, such as: effective flow, non-deformability, resistance, hydraulic seal, compliance of connection, correct opening / closing operation.

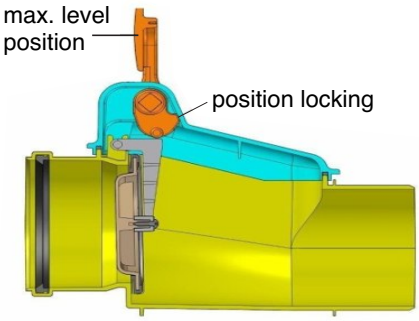
Backflow without the presence of NRV



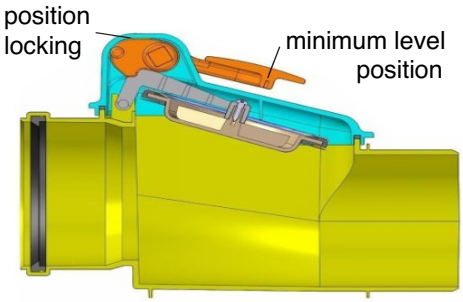
Backflow with the presence of NRV



Access pipes in a closed state - longitudinal section

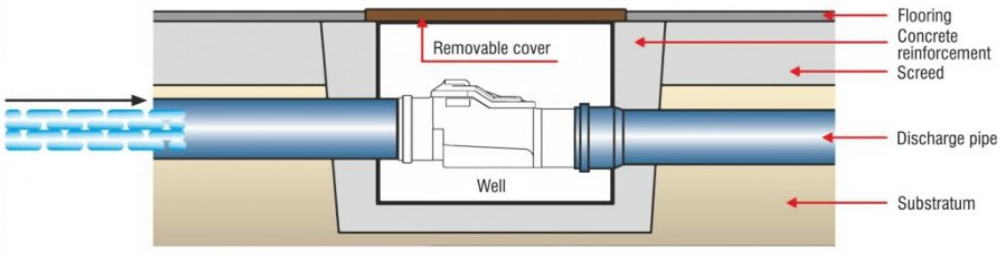


Access pipes in the open state - longitudinal section

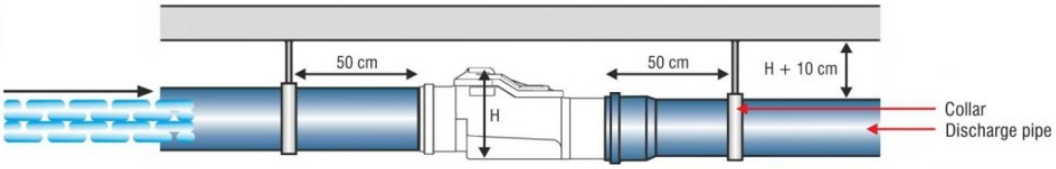


Installing the non-return valve

The non-return valve must be positioned upstream of the main manifold in a suitable accessible well

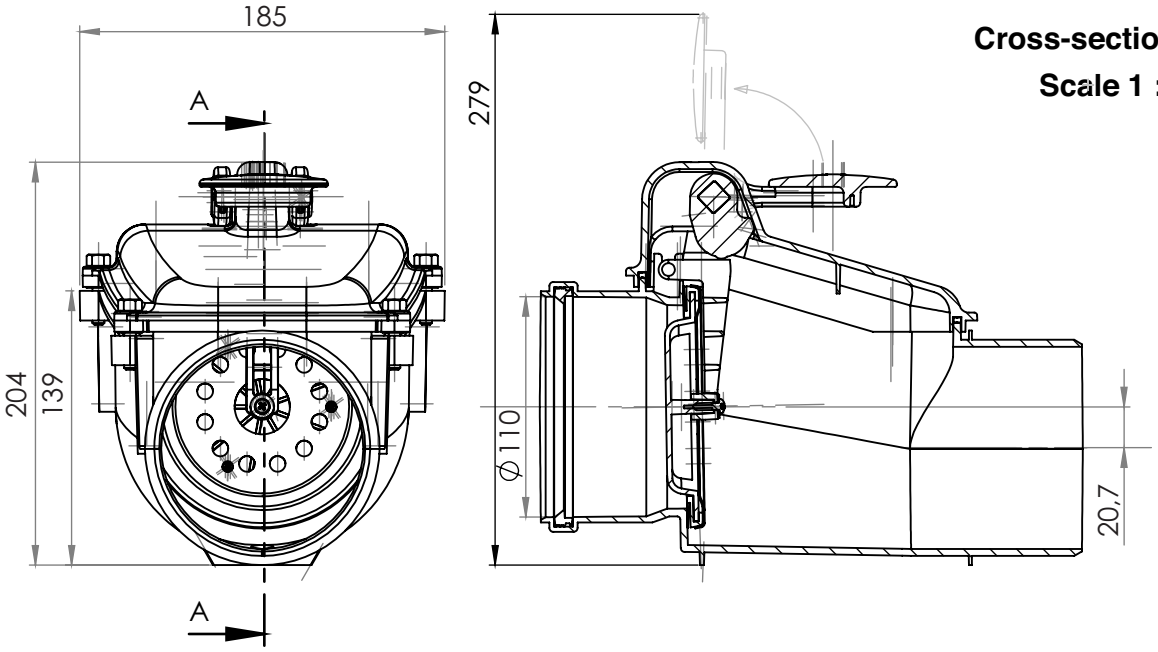


or by intercepting a pipe suspended horizontally on the intrados of the slab

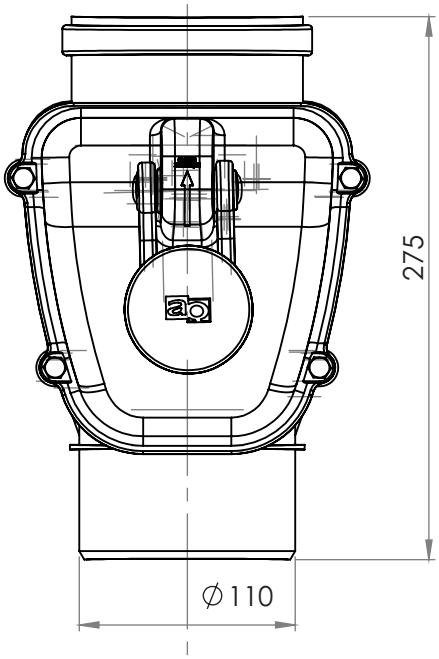


NON-RETURN VALVE
110 mm





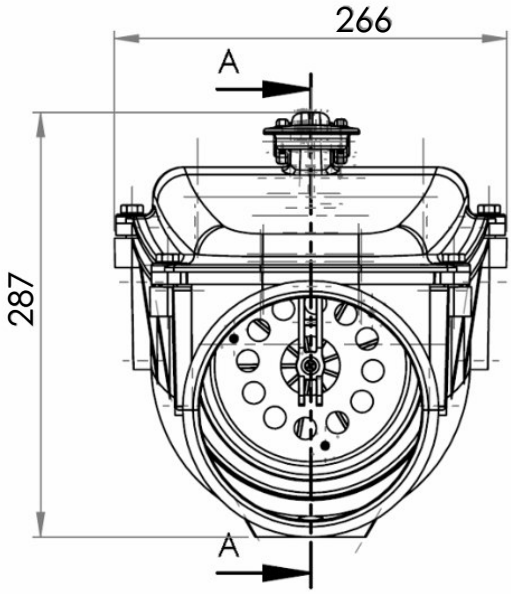
Cross-section A-A
Scale 1 : 3

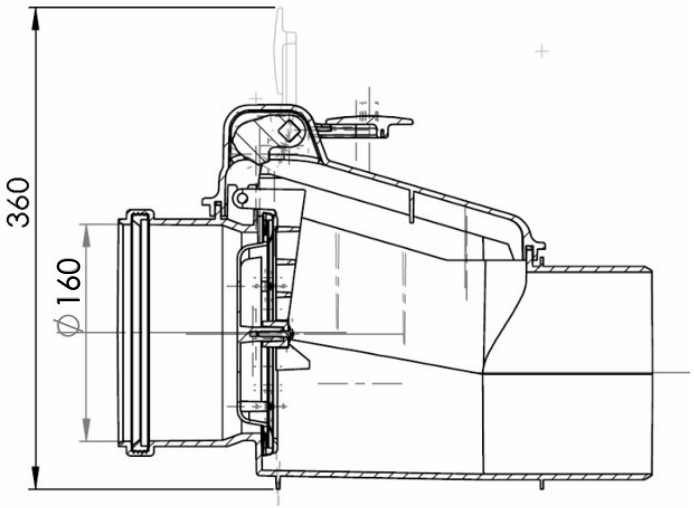


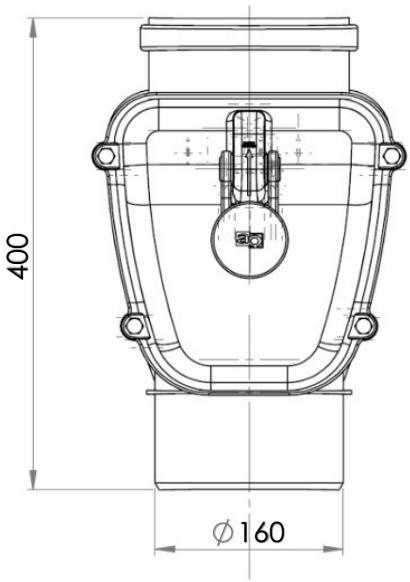
NON-RETURN VALVE
160 mm



Cross-section A-A

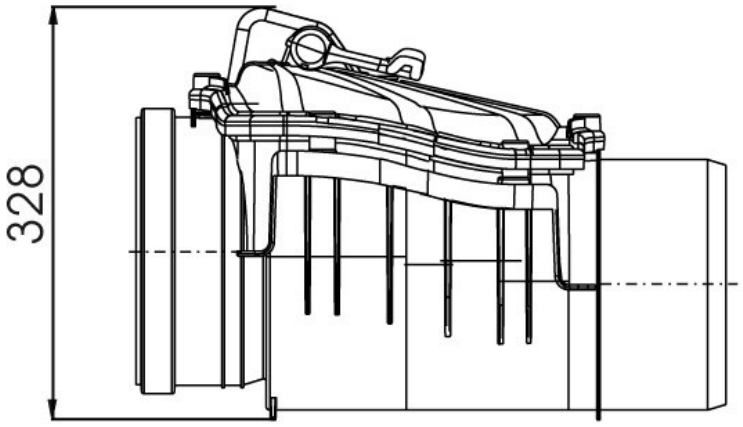


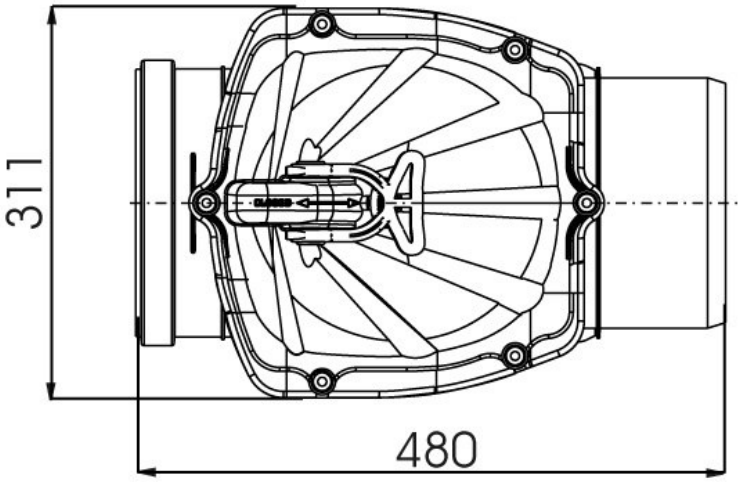




NON-RETURN VALVE
200 mm







NON RETURN VALVES DATA SHEET

1. Product name: Non return valves – diameters Ø110, Ø160, Ø200

2. Product classification: PKWiU 22.21.29.0

3. Intended use and scope: for use in non-pressure sewage (maximum working pressure: 0.5 bar), for installation on sewage drain pipes and thus not intended for installation on cooling and heating equipment.

4. Operation summary:

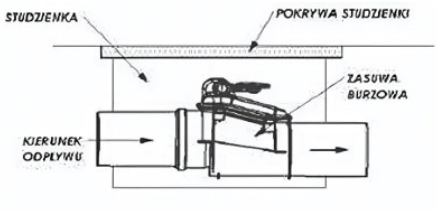
   4.1 **Installation:** The direction of flow must match the direction of the arrow on the valve lid.
   4.2 **Operation:** The check valve closes automatically upon sewage backflow. The valve lever switch must be in the "ON" position to allow drainage.
   4.3 **Emergency closure:** The "OFF" position of the lever switch prevents the flow of wastewater in both directions.
   4.4 **Maintenance:** Inspect the valve at least twice a year. During inspection or in cases of valve malfunction, remove the lid, check the sealing gaskets, clean and replace the damaged parts.

5. Reference Documents: EN 13564-2



NON RETURN VALVE OPERATION MANUAL

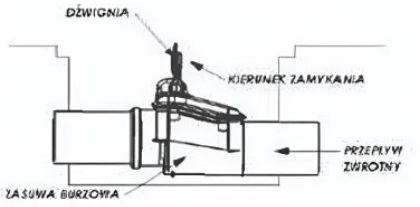
Installation



The non return valves are designed for installation in sewage systems running below the emergency sewage level of the external sewage system, within inspection manholes located at the sewage system building exits.

Install horizontally. For use in general access areas free of debris and/or liquid contaminants.

Usage

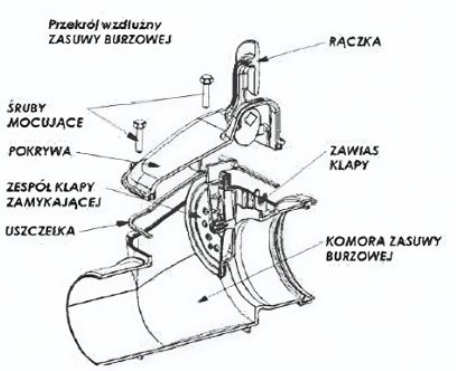


The non return valve is equipped with a locking switch.

Keep the lever in horizontal position to allow unidirectional flow of wastewater.

Set the lever to vertical position to completely shut off the flow.

Maintenance



Non return valves should be cleaned at least once every 6 months.

To clean the valve:

a. Clean the surface of the non return valve surface and the installation site prior to any other maintenance activities;
b. Unscrew each lid-securing bolt by diagonal half-turn
c. Remove the lid
d. Thoroughly clean the inside of the valve from dirt
e. Replace the sealing gaskets
f. Place the partition on the hinge
g. Put the lid on
h. Tighten each lid-securing bolt by diagonal half-turn
I. Check the flow shut-off mechanism

Non return valves: logistics data					
size (mm)	Weight [kg]		Dimensions [mm]		
	net	gross	height	width	length
110	0.785	0.930	210	190	280
160	2.150	2.400	290	270	412
200	4.200	4.550	340	310	490

PRODUCT DATA SHEET

Flowline fittings

Flowline is a complete system of pipes and fittings for the construction of outdoor sewage systems. Flowline fittings are produced by injection moulding and are factory-fitted with elastomer seals.

The wide range of types of Flowline fittings and their compatibility with other sewerage systems make it possible to make the most technically advanced sewage systems.



System name	Flowline fittings
Intended use	Rainwater, sanitary and combined sewer systems, sewer connections, sub-soil installations in buildings
Areas of application	UD
Material	Unplasticised polyvinyl chloride uPVC/PP Polypropylene
Colour	Orange- brown colour
Design of fittings	Solid homogeneous wall
Designation	SDR41
Diameter	DN/OD 110 mm, DN/OD 160 mm, DN/OD 200 mm, DN/OD 250 mm, DN/OD 315 mm, DN/OD 400 mm, DN/OD 500 mm
System components	Bends, branches, reducers, sliding couplings, two-pipe couplings, couplings for cast iron and vitrified clay pipes, plugs, caps, clean-outs, brick slips, storm gates
Circumferential stiffness classes	SN >= 8 kN/m^2 (EN ISO 13967:2019)

Chemical resistance	High chemical resistance to aggressive effluents and environments according to ISO/TR 10358 and ISO/TR 7620, pH 2-12
Maximum sewage temperature	DN110-500: 60°C (instantaneous effluent flow up to 5 min. up to 75°C)
Type of connections	Socket connections with elastomer seals
Can be connected to other systems	With DN/OD smooth-walled pipe systems, including the KG2000 PP-MD DN/OD system; with DN/OD socket sockets of SC manholes; with DN/ ID structural pipe systems – transition coupling DN OD/DN ID required
Can be used in areas of mining damage	Positive opinion of GIG up to and including mining damage class IV (DN110-DN315)
Standards, national technical specifications	EN 1401-1, EN 1329-1 National Institute of Hygiene hygienic certificate, Technical Opinion of GiG on the perimeter stiffness of uPVC fittings Flowline
Additional information	Replacement thermoplastic seals with stiffening ring (DIN-LOCK) possible

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- **Multan:** (92-61) 4545259
  E-mail: multan@dadex.com.pk
- **Islamabad:** (92-51) 5975109 - 110
  E-mail: islamabad@dadex.com.pk

For Technical information and assistance, please contact Dadex Techno Commercial Department at sales@dadex.com.pk

Note: All information contained in this literature is provided in good faith. The user should, however, verify that the product is fit for the intended purpose and application. Please ensure compliance with all health and safety requirements. While continuing its program of continuous development, the Seller reserves the right to modify or extend any published information without prior notification. No responsibility can be accepted for errors, omissions, or incorrect assumptions.

Seller's Responsibility: The responsibility of the seller ceases once the goods are delivered to the buyer's representative at our factory, where delivery is taken in person. In other cases, the responsibility of the seller shall cease once the goods are delivered to the buyer's or carrier's authorized representative(s). No claims of any type—including in-transit loss, damage, pilferage, or short-delivery—will be entertained by the seller, and the buyer agrees to hold the seller harmless in this regard. Additionally, the seller shall not be responsible for any consequential damages, including but not limited to economic loss of any kind whatsoever, upon the products being delivered to the buyer as per the terms of this Clause. Any claim or responsibility as stated herein will not be entertained by the seller, and such action shall not be the cause of a dispute by the buyer.

Jul-2026